WEEK-2

TASK: 1

```
├── .git
│   ├── HEAD
│   ├── config
│   ├── description
│   ├── hooks
│   │   ├── applypatch-msg.sample
│   │   ├── commit-msg.sample
│   │   ├── fsmonitor-watchman.sample
│   │   ├── post-update.sample
│   │   ├── pre-applypatch.sample
│   │   ├── pre-commit.sample
│   │   ├── pre-merge-commit.sample
│   │   ├── pre-push.sample
│   │   ├── pre-rebase.sample
│   │   ├── pre-receive.sample
│   │   ├── prepare-commit-msg.sample
│   │   ├── push-to-checkout.sample
│   │   ├── sendemail-validate.sample
│   │   └── update.sample
│   ├── info
│   │   └── exclude
│   ├── objects
│   │   ├── info
│   │   └── pack
│   └── refs
│       ├── heads
│       └── tags
├── .gitignore
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main
│   ├── main.c
│   ├── parser.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c
```

```
hphariprIya@LAPTOP-M4TV03SK:~/OSSP/src$ cat main.c
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
    }
    else if (pid == 0) {
        printf("This is the Child Process.\n");
        printf("Child PID = %d\n", getpid());
    }
    else {
        printf("This is the Parent Process.\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
```

```
hphariprIya@LAPTOP-M4TV03SK:~/OSSP/src$ ./main
This is the Parent Process.
This is the Child Process.
Parent PID = 1266
Child PID = 1267
Child PID = 1267
```

TASK:2

hpharipriya@LAPTOP-M4TV0

GNU nano 8.7.1 exec_demo.c

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
    }
    else if (pid == 0) {
        printf("Child Process\n");

        // Replace child process with "ls -l"
        execl("/bin/ls", "ls", "-l", NULL);

        // Only runs if exec fails
        printf("Exec failed!\n");
    }
    else {
        wait(NULL);
        printf("Parent Process Finished.\n");
    }

    return 0;
}
```

```
hpharipriya@LAPTOP-M4TV03SK:~/OSSP/src$ cd ~/OSSP/src
hpharipriya@LAPTOP-M4TV03SK:~/OSSP/src$ nano exec_demo.c
hpharipriya@LAPTOP-M4TV03SK:~/OSSP/src$ gcc exec_demo.c -o exec_demo
hpharipriya@LAPTOP-M4TV03SK:~/OSSP/src$ ./exec_demo
Child Process
total 56
-rwxr-xr-x 1 hpharipriya hpharipriya 16080 Aug  1 13:44 a.out
-rw-r--r-- 1 hpharipriya hpharipriya     0 Aug  1 12:55 builtin.c
-rwxr-xr-x 1 hpharipriya hpharipriya 16080 Aug  1 13:46 exec_demo
-rw-r--r-- 1 hpharipriya hpharipriya   519 Aug  1 13:42 exec_demo.c
-rw-r--r-- 1 hpharipriya hpharipriya     0 Aug  1 12:55 executor.c
-rwxr-xr-x 1 hpharipriya hpharipriya 16080 Aug  1 13:16 main
-rw-r--r-- 1 hpharipriya hpharipriya   444 Aug  1 12:59 main.c
-rw-r--r-- 1 hpharipriya hpharipriya     0 Aug  1 12:55 parser.c
-rw-r--r-- 1 hpharipriya hpharipriya     0 Aug  1 12:55 shell.c
Parent Process Finished.
hpharipriya@LAPTOP-M4TV03SK:~/OSSP/src$
```